ĐẠI HỌC QUỐC GIA TP. HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN

OPERATING SYSTEM

LAB 01 | INTRODUCTION TO LINUX

Introduce how to use the Ubuntu OS and discover different aspects of Operating System

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INSTRUCTOR

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OUTCOMES

After this LAB, students will be able to:

1. Efficiently use the Ubuntu OS by executing some basic shell commands
2. Describe the structure of a command in Linux
3. Clarify the path system in Linux
4. Describe the file permission concept in OS

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CONTENS

PREPARATION

1. Introduction to VSCode for CS50

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CONTENS

PRACTICE

1. Why Ubuntu?
2. What is shell?
3. Linux command’s structure
4. Path
5. Some basic commands
6. File mode

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INTRODUCTION TO VSCODE for CS50

PREPARATION

0.

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0. Introduction to CS50.dev system

Visual Studio Code for **CS50** is a web app at **cs50.dev** that adapts GitHub Codespaces for students and teachers.

It automates the process of creating a repository inside of CS50's GitHub organization, pushing to it an initial .devcontainer.json, and creating a "codespace," which is a Docker "container" in the cloud (akin to your very own virtual server).

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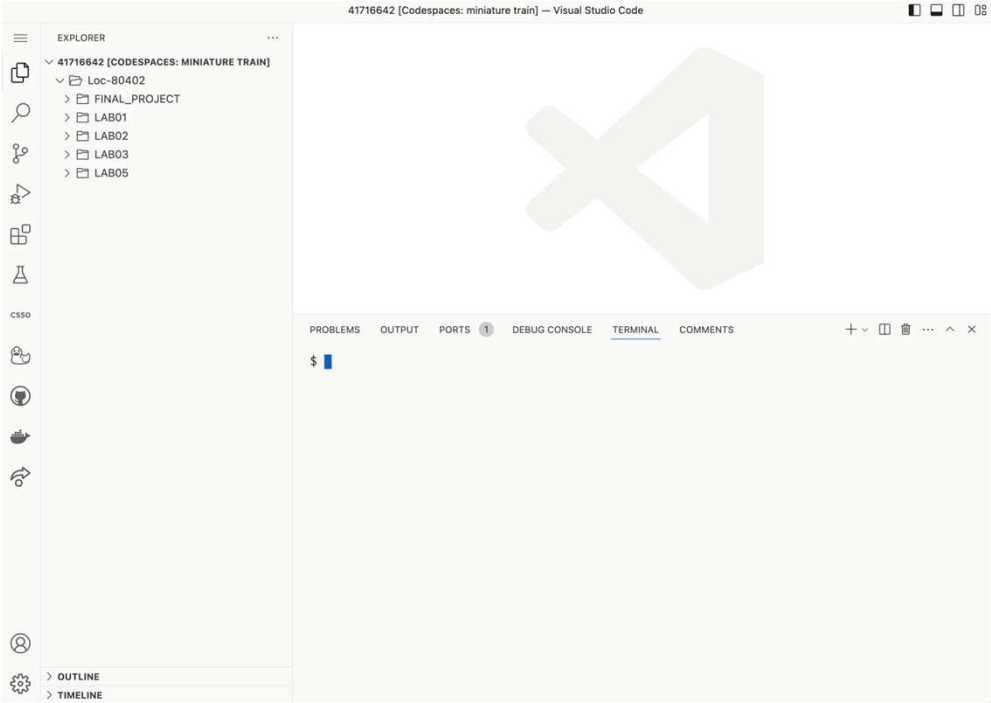
0. Introduction to CS50.dev system

Ultimately, it allows students to start programming with just a browser, without needing to install or configure anything locally on their own computer.



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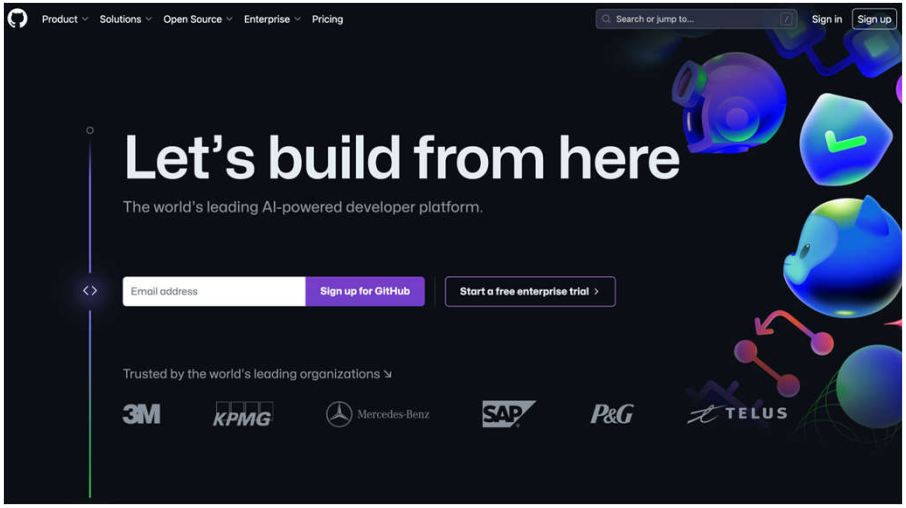
0. Introduction to CS50.dev system

- In order to use CS50.dev system, please first register a **github account** from the link below

Access: **github.com**



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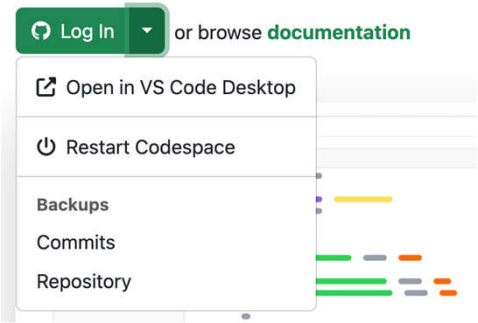
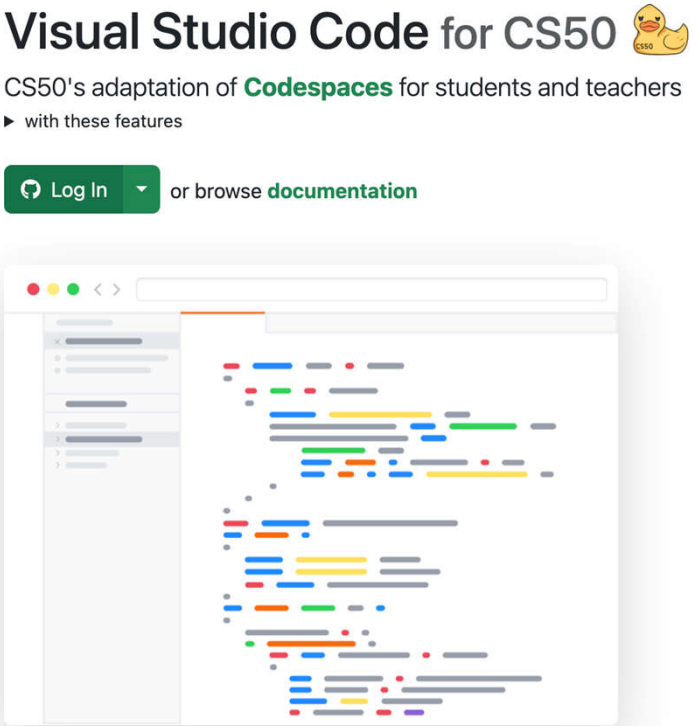
0. Introduction to CS50.dev system

- Use your github account to log in the CS50.dev Virtual Machine

Access: **cs50.dev**



You can even use CS50.dev in your **VSCode Desktop**



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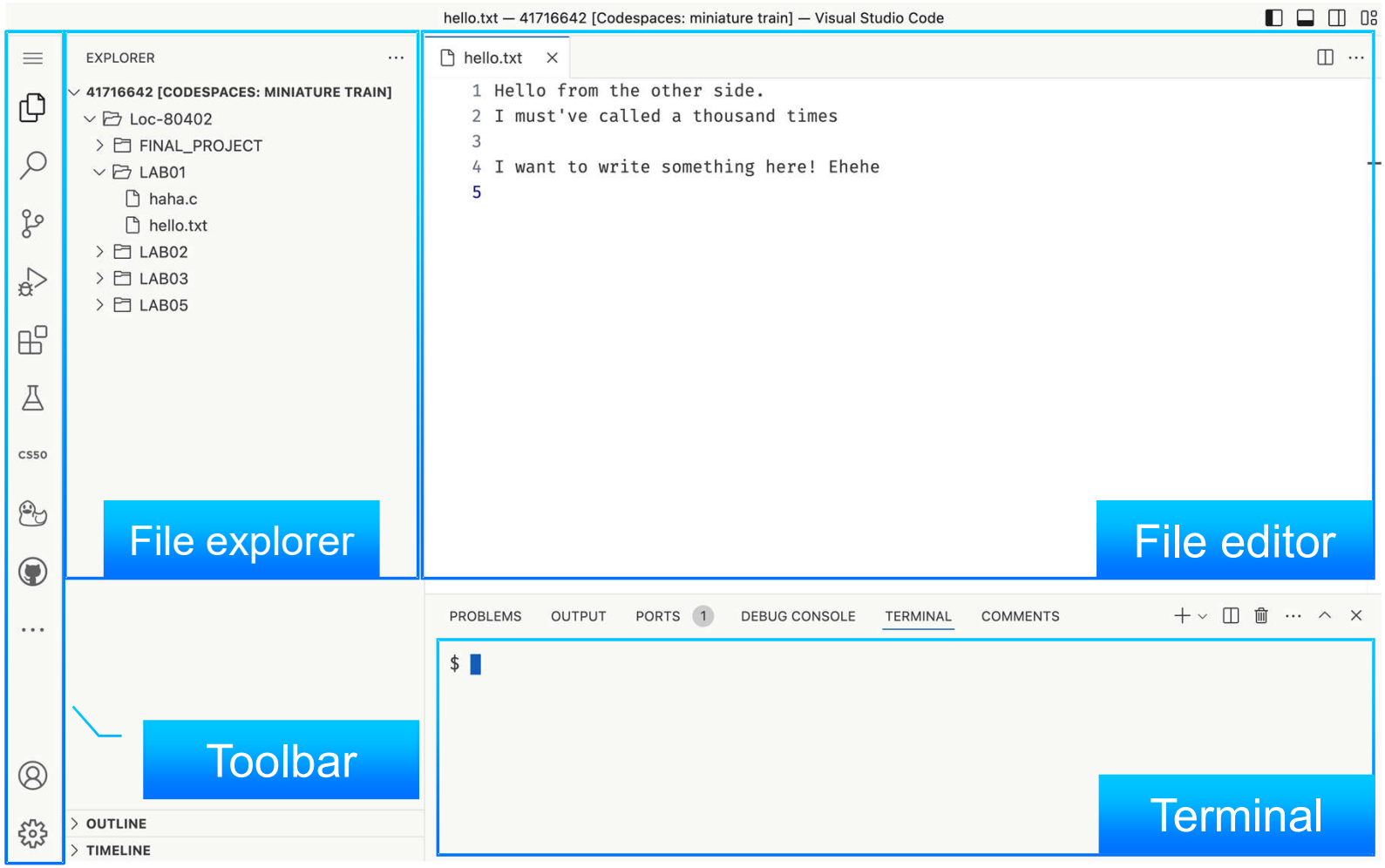
0. Introduction to CS50.dev system



0. Introduction to CS50.dev system

Toolbar

- File explorer
- Search
- Source control (git)
- Debug
- Extension
- CS50 menu
- CS50 Duck Debugger
- Github Pull Request
- Docker
- Live Share

- Accounts
- Manage

File explorer

Contains files and folders and your workspace

File editor

Where you can read and edit files chosen in File explorer

Terminal

Where you can interact with the VM by using shell commands

0. Introduction to CS50.dev system

- When using Visual Studio Code for CS50 (CS50.dev), you remotely connect to a virtual server with the help of Github codespace and Docker container
- The virtual server is running Ubuntu

0. Introduction to CS50.dev system

Classwork: 01

Try to find out the the computer hardware configuration and Ubuntu version of the VM by executing beside commands in the Terminal. Explain the use of each command.

```
sudo unminimize
sudo apt-get install lshw
sudo lshw
df -h
cat /etc/*release
```

0. Introduction to CS50.dev system

HARDWARE	OPERATING SYSTEM
CPU: Memory:	OS:

WHY UBUNTU

01.

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1. Why Ubuntu?

1. **Open source nature**
   - Access to Source Code, Learning Opportunity
2. **Understanding core concepts**
   - Hands-On Experience, Customization and Control
3. **Scripting and automation**
   - Shell Scripting, Scriptable Environment
4. **Server and Networking skills**
   - Real-World Usage, System Administration

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1. Why Ubuntu?

5. Security and Stability

- Security Practices, Stability and Reliability

6. Career and Industry relevance

- Demand in Tech Industry, Certifications and Opportunities

7. Community and Support

- Strong Community, Collaborative Learning

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WHAT IS SHELL?

02.

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2. What is Shell?

A shell is a program that **takes commands from the user** and **gives them to the operating system to perform**. It's a bridge between the user and the computer's underlying system.

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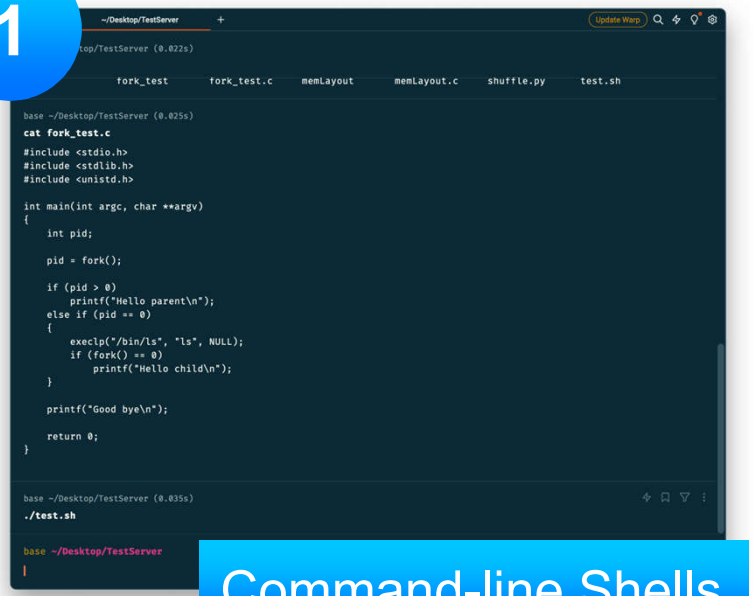
2.1. Types of shell

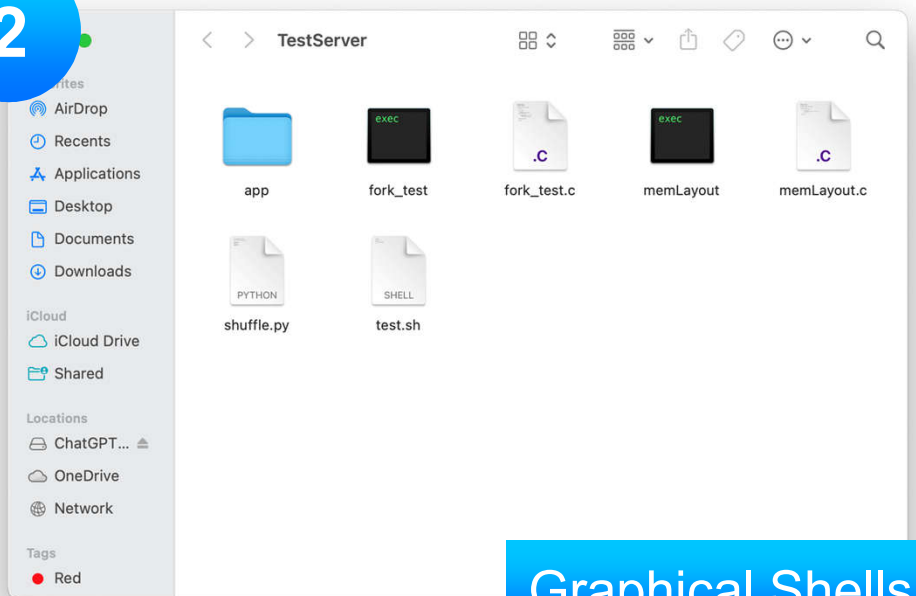
There are primarily two types of shells

1



Command-line Shells

2



Graphical Shells

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2.2. How shells work?

INPUT

Users type commands or use the GUI to issue commands

PROCESSING

The shell interprets these commands and communicates them to the operating system

OUTPUT

The shell displays the results of these commands back to the user

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2.3. Common tasks

- **File management**

  Creating, deleting, moving, and copying files and directories

- **Program execution**

  Running software programs and scripts

- **System monitoring**

  Checking system performance and resource usage

- **Configuration**

  Changing system settings and configurations

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2.4. Examples

Bash commands	Powershell commands
• **ls**: Lists files and directories	• **Get-ChildItem**: Lists files and directories.
• **cd**: Changes the current directory	• **Set-Location**: Changes the current directory.
• **cp**: Copies files or directories	• **Copy-Item**: Copies files or directories.

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2.5. Benefits of using a Shell

- **Efficiency**

  Advanced users can perform complex tasks quickly through commands and scripts

- **Automation**

  Scripts can automate repetitive tasks, saving time and reducing errors

- **Flexibility**

  Shells often provide powerful tools and utilities that are not available through graphical interfaces

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LINUX COMMAND’s STRUCTURE

03.

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3. Linux Command's Structure

command params -options

command: name of command, indicate **what** user want to do

params: parameters which are used by command (in many cases, why also called it command-line params)
optional

-options: indicates **how** command is executed
optional

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3. Linux Command's Structure

Examples

ls	# only command
ls ~/Loc-80402	# command + 1 param
ls ~/Loc-80402 –laht	# command + 1 param + 4 options

Note:

- Each part in command is **separated** by **1 (one) space**
- Some commands may have **multiple params** and **options**
- Run **man <command>** to read **instruction** of that command

PATH

04.

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4. Path

You can use command **cd** to **c**hange the current **d**irectory

cd path

Examples

```
cd /usr/local/bin        # path: /usr/local/bin
cd /workspaces           # path: /workspaces
cd 41716642              # path: 41716642
                         # change 41716642 into your VM ID
```

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4.1. What is path?

A path is a string of characters that **specifies the location** of a file or directory within a file system. It acts as an **address**, guiding the operating system to the desired file or directory.

Two types of path:

- Absolute path
- Relative path

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4.2. Absolute path

- An absolute path provides the complete address to a file or directory, starting from the root directory.
- It uniquely identifies a file or directory's location in the entire file system, regardless of the current working directory.

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4.2. Absolute path

- Always starts from the root directory
- Always begins with a root directory symbol
- Provides a complete and unambiguous path

Examples

/usr/local/bin

/workspaces/4176642

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4.2. Relative path

- A relative path specifies a file or directory location in relation to the current working directory.
- It does not start from the root directory but instead is based on the current directory from which the command or script is executed.

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4.2. Absolute path

- Does not start with a root directory symbol
- Depends on the current working directory
- Can use special symbols to navigate:
  - . : represent *current* directory
  - .. : represent *parent* directory
  - ~ : represent *home* directory

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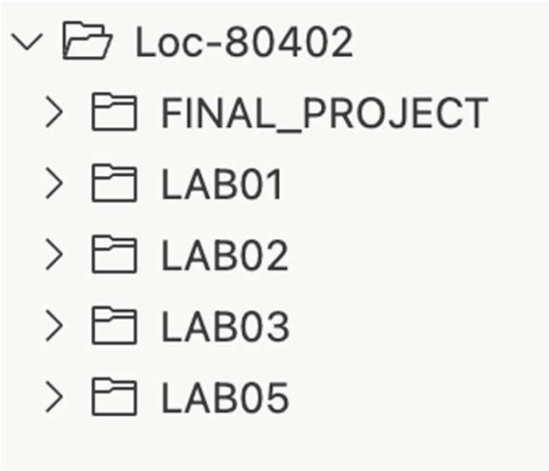
4.2. Absolute path



Examples

*Assume that we are in **LAB01***

```
cd ..          # Go to Loc-80402
cd LAB02       # Go to LAB02
cd ../LAB03    # Go to LAB03
ls .           # List items in LAB03
ls             # List items in LAB03
```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SOME BASIC COMMANDS

05.

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5. Some basic commands

Classwork: 02

Execute, google, and explain usage of all commands in this section

ls

cd

pwd

touch

nano

mkdir

rm

cp

mv

echo

cat

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FILE MODE

06.

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6.1. File mode

THREE TYPES OF USER

Name	Role	Symbol
User/Owner	owner of the file	u
Group	People in the same group with owner	g
Other	other people	o

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6.1. File mode

THREE TYPES OF PERMISSION

Name	Number	Symbol
Read	4	r
Write	2	w
Execute	1	x

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6.1. File mode

```
$ ls test.txt -l
-rw-r--r-- 1 ubuntu ubuntu 1 Jun 18 14:45 test.txt
```

Directory?

Other's permission

Group's permission

Owner's permission

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6. Change mode

You can change file mode by using

chmod <mode> file

761

mode is the sum of permission represented by number

first number: permission of owner

second number: permission of group

third number: permission of other

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6. Change mode

You can change file mode by using

chmod <mode> file

Examples

chmod 777 test

chmod 664 hello.txt

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6. Change mode

You can change file mode by using

chmod <role operator permission> file

Operator	Meaning
+	Add permission
-	Remove permission
=	Set permission

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6. Change mode

You can change file mode by using

chmod <role operator permission> file

Examples

chmod u+w test

chmod u-x,g+w,o=rw hello.txt

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EXERCISES

07.

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Classwork: 03

Create a folder tree like below image:

```
Myweb
    |-images
    |      |-- icon
    |      |-- background
    |      |-- animation
    |-databases
    |-scripts
    |-java
```

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Classwork: 04

Google the way to find some files with “.html” extension in your entire system. Copy one of them to Myweb.

Classwork: 05

Remove the directory Myweb/scripts.

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Classwork: 06

Set the permision for Myweb/databases as below:

- The owner has all permission
- Other users have no permission

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HOMEWORK

08.

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Homework A

Show the content of the text file one page at a time using the **less** command.

Homework B

Use **grep** to search your file for lines containing a word of your choice

Homework C

Show the first and last ten lines of the file using **head** and **tail**, respectively.

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